ASSIGNMENT 1 REPORT

K-Nearest Neighbors (KNN) Classification

Group Number: 6

Group Members

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This report is based on the submitted Group06_Assignment1.ipynb notebook. Figures and numerical outputs included here are taken from the notebook's recorded outputs.

1. Assignment Requirements

The assignment requires KNN classification from scratch for a binary dataset and for CIFAR-10 multiclass classification. The required K values are 3, 4, 9, 20 and 47, and the required distance measures are Euclidean, Manhattan, Minkowski, Cosine and Hamming. For each task, the best configuration is selected using testing accuracy, followed by confusion matrix, precision, recall, accuracy-vs-K analysis and observations. The bonus asks for a decision-boundary visualization. The assignment explicitly states that pre-built model functions from libraries such as sklearn or pytorch should not be used. ■filecite■turn6file0■L52-L66■ ■filecite■turn6file0■L73-L90■

2. Task 1 — Dataset and Preprocessing

The notebook loads data.csv and reports 569 samples with 30 features. The target is Diagnosis, where M represents malignant and B represents benign. The data are shuffled using random seed 42 and divided into 455 training samples and 114 testing samples, corresponding to an 80:20 split. Standardization is performed using training-set statistics only.

3. Task 1 — KNN Experiments

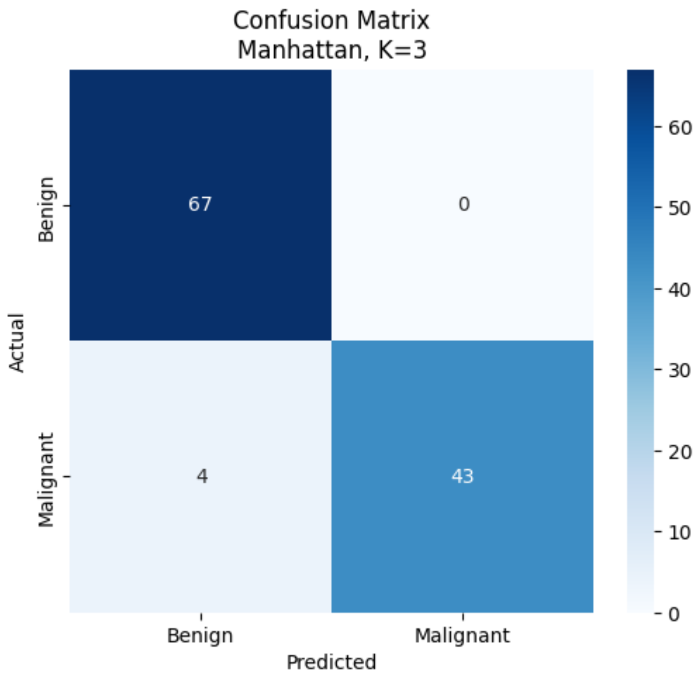
Twenty-five experiments are performed: five distance measures × five K values.

Distance	K	Accuracy
Euclidean	3	0.9474
Euclidean	4	0.9298
Euclidean	9	0.9298
Euclidean	20	0.9386
Euclidean	47	0.9386
Manhattan	3	0.9649
Manhattan	4	0.9386
Manhattan	9	0.9474
Manhattan	20	0.9386
Manhattan	47	0.9386
Minkowski	3	0.9474
Minkowski	4	0.9298
Minkowski	9	0.9298
Minkowski	20	0.9386
Minkowski	47	0.9386
Cosine	3	0.9649
Cosine	4	0.9561
Cosine	9	0.9474
Cosine	20	0.9561
Cosine	47	0.9561
Hamming	3	0.7105
Hamming	4	0.6228
Hamming	9	0.6404
Hamming	20	0.6667
Hamming	47	0.5877

Best Task 1 result: Manhattan distance, K=3, accuracy 0.964912 (96.49%). Cosine distance with K=3 also gives 0.964912.

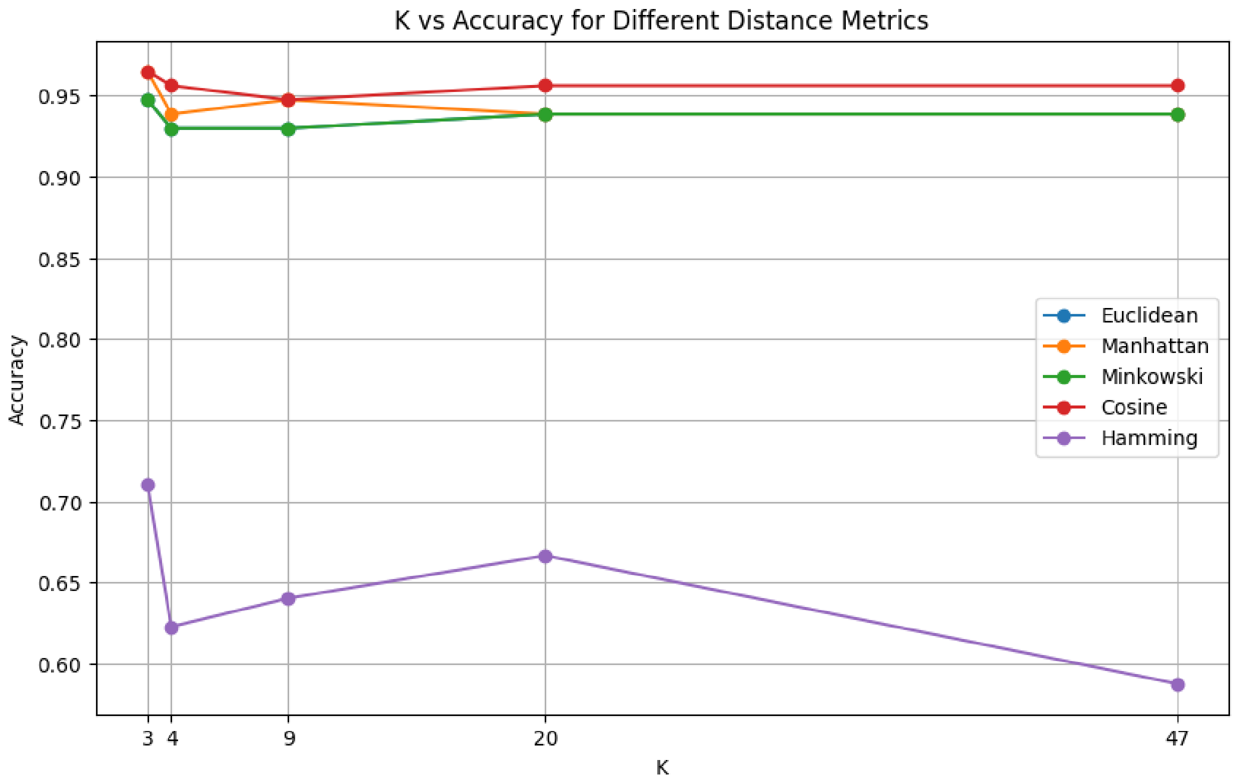
4. Task 1 — Confusion Matrix, Precision and Recall

For the selected Manhattan K=3 model, the notebook reports the confusion matrix [[67, 0], [4, 43]], precision = 1.0000, recall = 0.9148936 and accuracy = 0.9649123.



The matrix shows no false positives and four false negatives. Therefore, the malignant-class precision is perfect in this test set, while recall is reduced by the four missed malignant samples.

5. Task 1 — Accuracy Plot



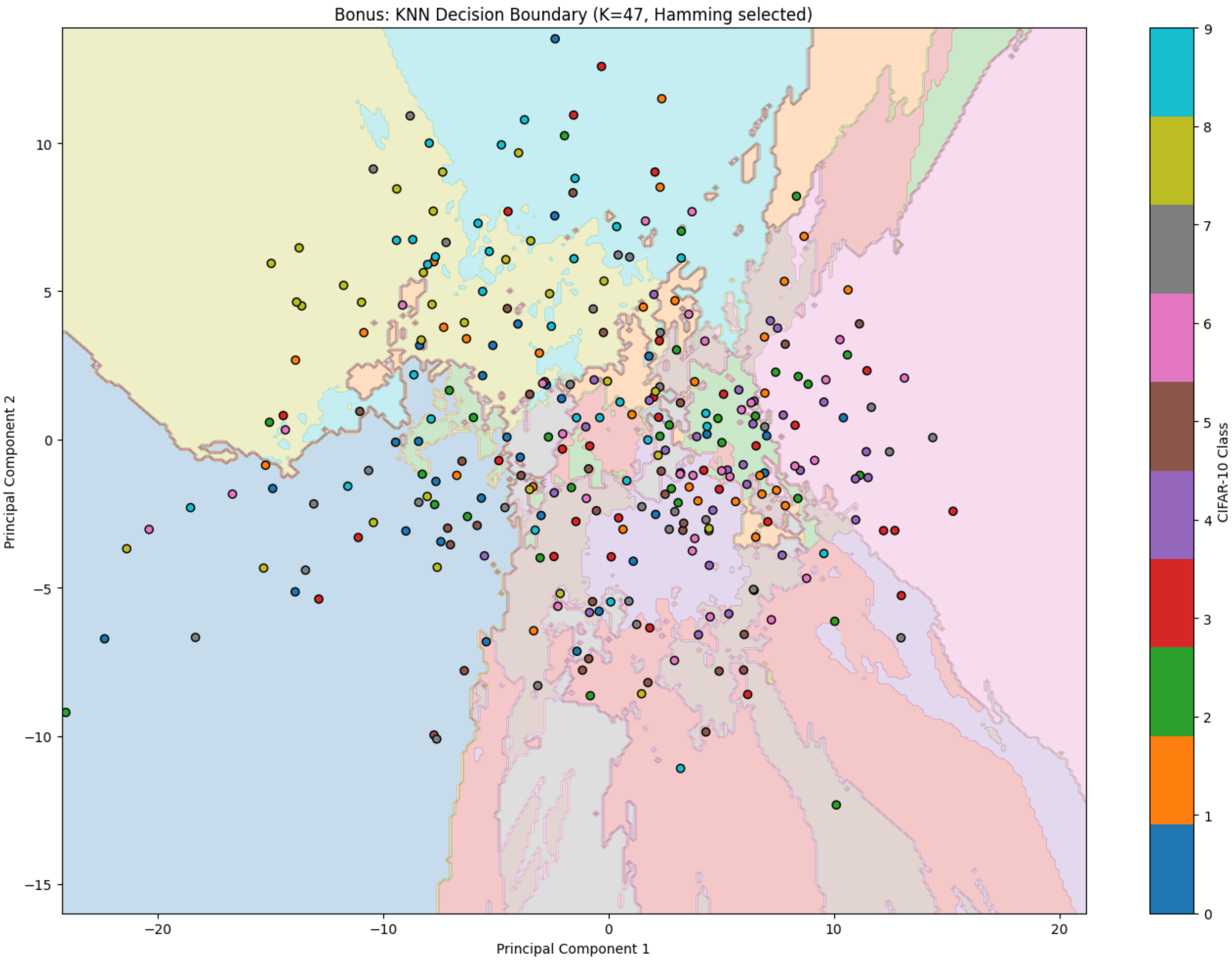
Observation: Manhattan and Cosine reach the highest Task 1 accuracy at K=3. Hamming is substantially weaker on these standardized continuous features.

6. Task 1 — Final Output

The notebook's final summary confirms: Best Distance = Manhattan, Best K = 3, Accuracy = 0.9649123, Precision = 1.0 and Recall = 0.9148936.

7. Bonus — Decision Boundary

The notebook includes the required bonus visualization. For this visualization, 1,000 CIFAR-10 training samples and 300 testing samples are reduced from 3,072 dimensions to two principal components. A K=47 KNN decision boundary is calculated over 62,500 grid points. The recorded 2D KNN accuracy is 0.1667. This is a PCA-based visualization experiment and is separate from the full-dimensional Task 2 accuracy.



8. Task 2 — CIFAR-10 Dataset

The notebook loads the CIFAR-10 folder structure successfully. It reports 50,000 training images and 10,000 test images, with each flattened image represented by 3,072 features. The ten classes are airplane, automobile, bird, cat, deer, dog, frog, horse, ship and truck.

To make the from-scratch KNN experiments computationally practical while preserving class balance, the notebook selects 500 training images and 100 testing images per class. The resulting balanced subset is 5,000 training images and 1,000 testing images, with exactly 500 training and 100 testing samples for every class.

A quick KNN test reports accuracy 0.60 on a smaller test batch in 0.64 seconds. The full experiment then runs 25 configurations.

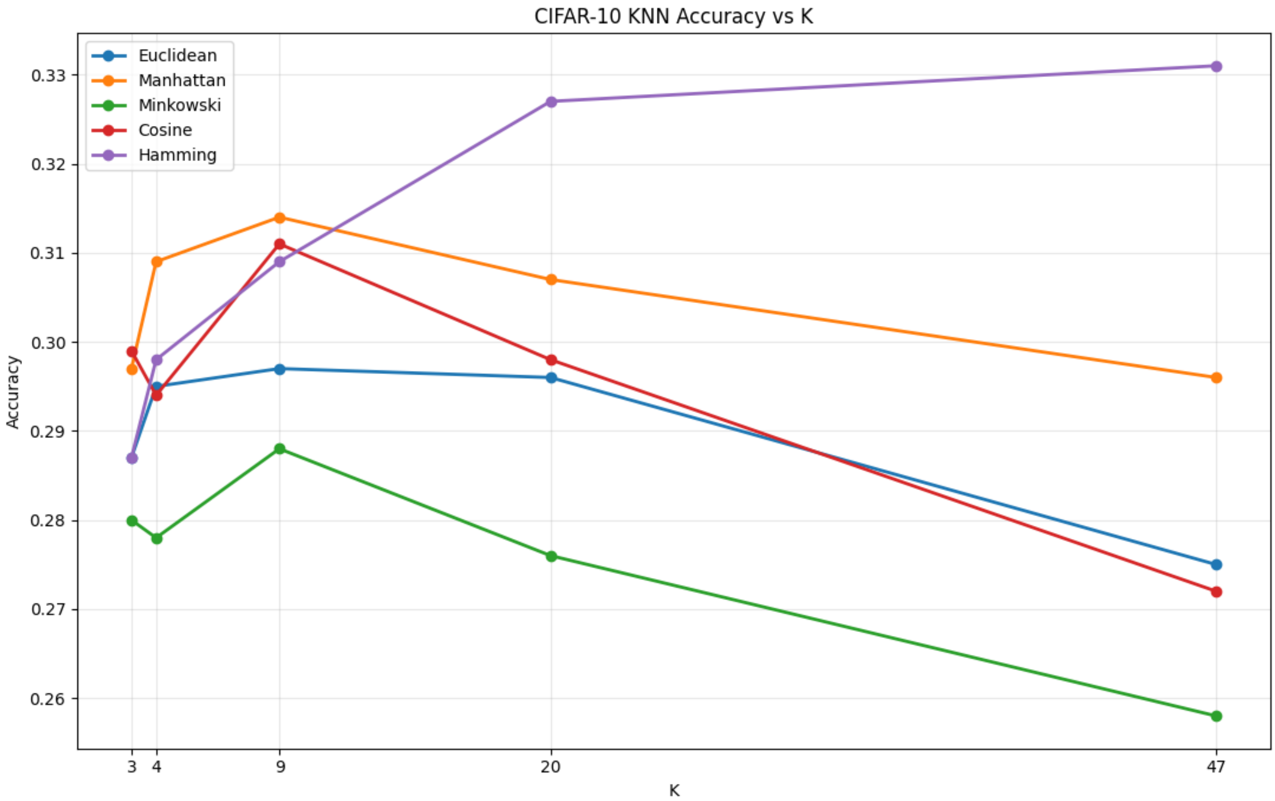
9. Task 2 — Experimental Results

Rank	Distance	K	Accuracy	Time (s)
1	Hamming	47	0.331	19.82
2	Hamming	20	0.327	20.85
3	Manhattan	9	0.314	49.25
4	Cosine	9	0.311	6.08
5	Manhattan	4	0.309	49.55
6	Hamming	9	0.309	20.57
7	Manhattan	20	0.307	49.14
8	Cosine	3	0.299	5.74
9	Cosine	20	0.298	4.69
10	Hamming	4	0.298	20.01
11	Manhattan	3	0.297	48.19
12	Euclidean	9	0.297	4.77
13	Euclidean	20	0.296	5.62
14	Manhattan	47	0.296	48.40
15	Euclidean	4	0.295	6.22
16	Cosine	4	0.294	4.75
17	Minkowski	9	0.288	295.93
18	Hamming	3	0.287	20.84
19	Euclidean	3	0.287	4.76
20	Minkowski	3	0.280	311.69
21	Minkowski	4	0.278	290.70
22	Minkowski	20	0.276	292.22
23	Euclidean	47	0.275	5.37
24	Cosine	47	0.272	4.72
25	Minkowski	47	0.258	290.79

Best Task 2 result: Hamming distance, K=47, accuracy 0.3310 (33.10%).

10. Task 2 — Confusion Matrix

The notebook generates and displays a 10×10 confusion matrix for the best Hamming K=47 model. The actual plotted figure is included below exactly as recorded in the notebook.

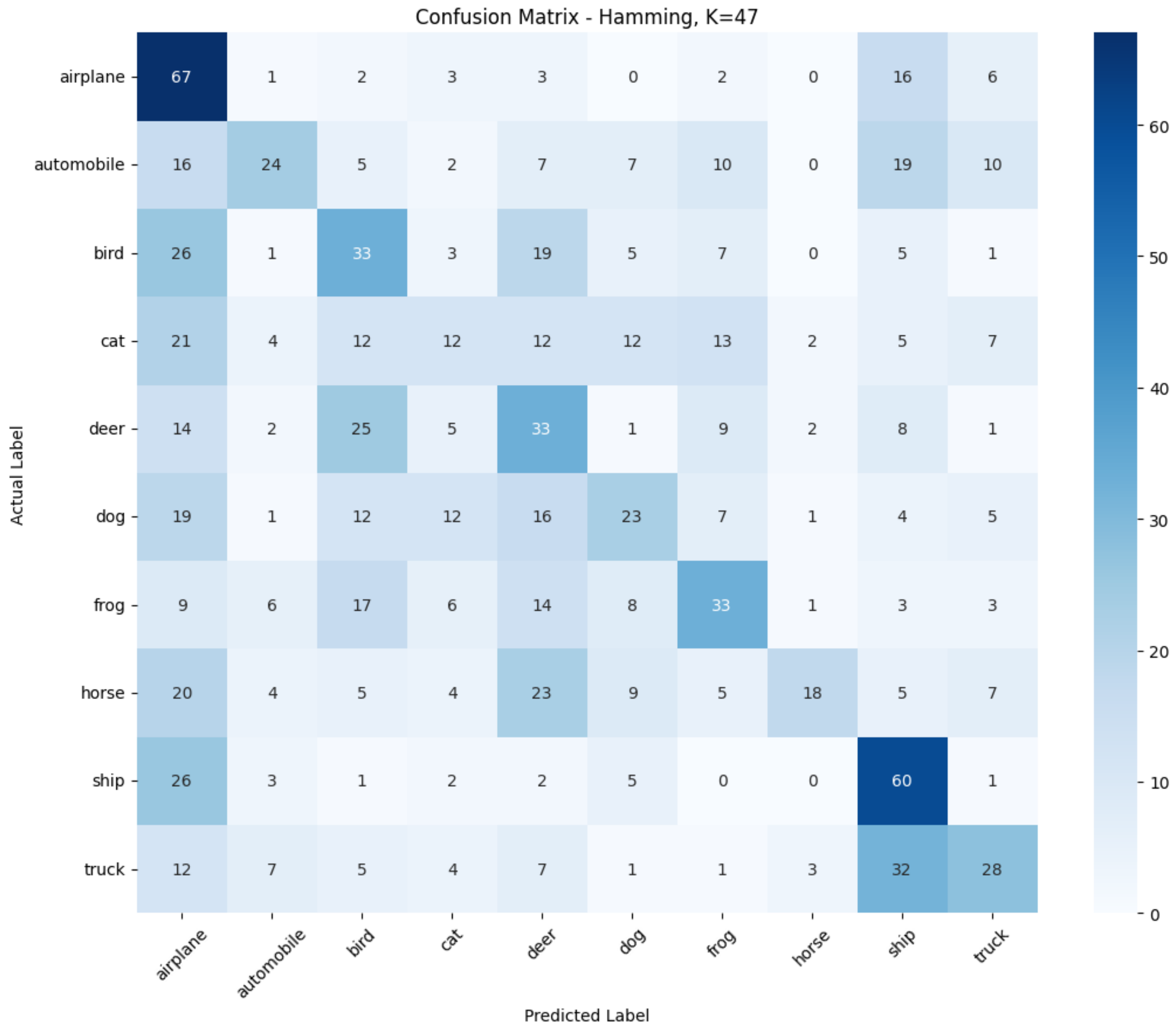


The confusion matrix is used to identify which CIFAR-10 classes are correctly classified and which classes are most frequently confused with one another.

11. Task 2 — Precision and Recall

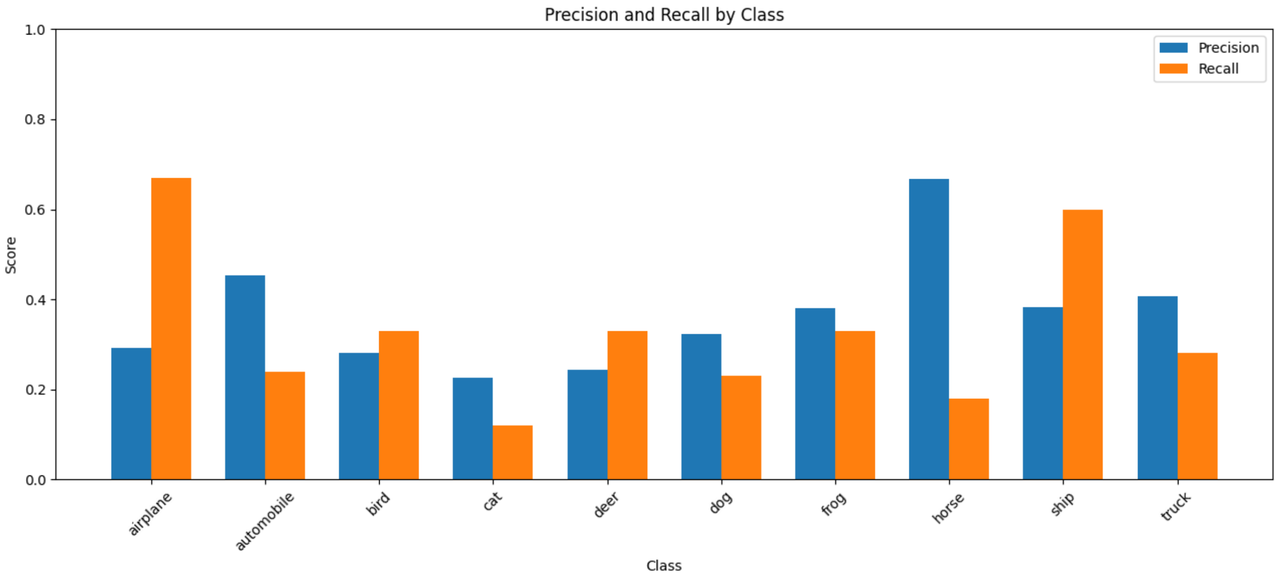
Class	Precision	Recall
airplane	0.2913	0.67
automobile	0.4528	0.24
bird	0.2821	0.33
cat	0.2264	0.12
deer	0.2426	0.33
dog	0.3239	0.23
frog	0.3793	0.33
horse	0.6667	0.18
ship	0.3822	0.60
truck	0.4058	0.28

Macro Precision = 0.3653; Macro Recall = 0.3310; Accuracy = 0.3310.



The class with the highest recall is airplane (0.67), followed by ship (0.60). Cat has the lowest recall (0.12). Horse has the highest precision (0.6667), while cat has the lowest precision (0.2264).

12. Task 2 — Accuracy Across K and Metrics



The plot shows the effect of K for all five distance measures. Hamming reaches the best overall result at K=47. Manhattan peaks at K=9 with 31.4%, while Cosine peaks at K=9 with 31.1%. Minkowski is consistently slower and its accuracy remains below the other metrics in these experiments.

13. Task 2 — Final Summary Output

Training images: 5,000; Testing images: 1,000; Number of experiments: 25; Best distance: Hamming; Best K: 47; Best accuracy: 0.3310; Macro precision: 0.3653; Macro recall: 0.3310. The notebook also records a final prediction time of 0.33 minutes for the best model.

14. Comparative Analysis

Task 1 is substantially easier than Task 2 in terms of classification accuracy. The best Task 1 configuration reaches 96.49%, while the best balanced-subset CIFAR-10 configuration reaches 33.10%. Task 1 uses 30 standardized numerical features and two classes. Task 2 uses 3,072-dimensional image representations and ten classes, making nearest-neighbour comparisons more difficult.

For Task 1, Manhattan and Cosine perform best at K=3. Hamming performs much worse on the standardized continuous features. For Task 2, Hamming performs best, reaching 33.1% at K=47. The results show that the most suitable distance metric depends strongly on the representation and dataset.

The timing output also demonstrates the computational cost of from-scratch KNN. Minkowski is the slowest metric in the Task 2 experiments, taking roughly 291–312 seconds per configuration, while Euclidean and Cosine are much faster on the balanced subset.

15. Conclusion

The notebook completes the required KNN experiments for both binary and multiclass classification, including all five specified distance measures and all five specified K values. Task 1 achieves 96.49% accuracy with Manhattan K=3. Task 2 achieves 33.10% accuracy with Hamming K=47 on the balanced CIFAR-10 subset. Confusion matrices, precision, recall, accuracy plots and the bonus PCA-based decision-boundary visualization are included. The experiments demonstrate how distance choice, K, dimensionality and the number of classes affect KNN performance.

16. Submission Files

The code notebook is Group06_Assignment1.ipynb and this report is the corresponding PDF report for Group 6.